

PII REDACTION TOOL

Evaluation Strategy & Metrics Report

Target Format: OpenXML (.docx) Document: Red Herring Prospectus.docxDate: 2026-08-14Status: PROVISIONAL BENCHMARK

Total Workload: 3,507 CandidatesAutomated Reviewed: 63 CandidatesHuman Reviewed: 0 CandidatesRegression Suite: 83 Passed

1. Objective

The primary objective of the PII Redaction Tool evaluation framework is to quantitatively measure the detection accuracy, structural preservation, pseudonymization integrity, and residual leakage of sensitive Personally Identifiable Information (PII) within complex OpenXML (.docx) financial prospectuses.

Financial prospectuses contain dense combinations of personal executive names, residential/registered addresses, contact email addresses, telephone numbers, and banking details interleaved with corporate financial tables and legal disclosures. The PII Redaction Tool parses document OpenXML structures across body paragraphs, table cells, running headers, and footers, detects candidate PII entities across 9 supported categories, deterministically maps sensitive entities to synthetic pseudonym replacements, applies run-level XML text substitution without corrupting document layout, and validates output documents for zero residual PII leakage.

2. Supported PII Categories

The evaluation framework natively covers 9 distinct PII categories:

PII Category	Inclusion Scope & Description	Detection Logic
PERSON	Names of natural persons (directors, promoters, officers)	Local spaCy NER + Honorific rules
EMAIL	Personal and corporate contact email addresses	Compiled regular expressions
PHONE	Indian mobile, landline, and formatted phone numbers	Compiled regex + STD area triggers
ORGANIZATION	Private corporate entity names (issuers, auditors, banks)	Local spaCy NER + Legal suffix rules
ADDRESS	Physical premises, registered offices, locality, PIN code	Multi-signal premises/PIN scoring
SSN	US Social Security Numbers (NNN-NN-NNNN)	Area/Group/Serial structured regex
CREDIT_CARD	Financial credit/debit card numbers (13-19 digits)	Candidate extraction + Luhn checksum
DOB	Dates of birth of natural persons	Contextual prefix triggers (DOB:, Born on)
IP_ADDRESS	IPv4 network addresses (192.168.1.1)	Lookaround bounds + ipaddress validation

3. Detection Approach

- The detection engine combines statistical Natural Language Processing (NLP) with deterministic rule-based algorithms:
- Local spaCy Statistical NER: Uses en_core_web_sm locally (zero cloud API dependencies) for PERSON and ORGANIZATION candidates.
 - Compiled Regular Expressions: Structured patterns (EMAIL, PHONE, SSN, CREDIT_CARD, IP_ADDRESS) use lookaround bounds to prevent partial figure matches.
 - Contextual Trigger Verification: DOBDetector requires explicit birth context (e.g. 'Date of Birth:'), preventing incorporation dates from being misclassified.
 - Algorithmic Validation Filters: CreditCardDetector validates candidates against the Luhn algorithm (_validate_luhn); IPDetector uses Python's ipaddress module.
 - Priority & Overlap Resolution: Resolves overlaps via category priority (EMAIL/PHONE/SSN/CC/IP > DOB > ADDRESS > ORGANIZATION > PERSON) and span length.

4. Redaction Approach

- Redaction is executed through a multi-pass pipeline:
- OpenXML Coordinates: DOCXExtractor tracks paragraph, table cell, row, and header coordinates (SourceLocation).
 - Co-Reference Propagation: Multi-pass scanning re-scans text chunks for known entity texts to capture repeated occurrences document-wide.
 - Deterministic Pseudonymization: Pseudonymizer generates MD5 seeds from (entity_type, text) to map recurring names to identical fake data.
 - Run-Level Text Substitution: DOCXRedactor substitutes text at the run level (<w:r>), preserving font styles, alignments, and borders.

5. CareEdge Bug Fix & Regression: A propagation flaw in repeated occurrence detection was fixed and verified via test_careedge_leak_regression.

5. Ground-Truth / Annotation Methodology

The evaluation dataset was constructed using an automated candidate generation and policy-based review workflow:

- Total Workload Candidates: 3,507
- Automated Provisional Reviewed: 63
- Independent Human Reviewed: 0
- Unreviewed Candidates: 3,444
- Accepted Provisional Entities: 15
- Rejected Provisional Candidates: 48
- Corrected Provisional Spans: 1 (cand_59)

PROVENANCE DISCLAIMER

Zero (0) candidates were independently reviewed by a human auditor. The 63 reviewed decisions were generated programmatically via automated policy simulation and constitute a provisional automated annotation benchmark. They must NOT be described as human-verified, human-reviewed, or human-audited gold-standard ground truth.

6. Entity-Level Matching

Entity matching incorporates both metadata and spatial coordinates:

1. Entity Type: Candidate entity_type must match ground-truth entity_type.
2. Spatial Coordinate Key: The unique SourceLocation (source_type, table_idx, row_idx, cell_idx, para_idx, header_idx, footer_idx) must match.
3. Character Span: Character offsets (start_offset, end_offset) within the source text chunk must match.

Why Coordinate Keys Are Mandatory: Identical text snippets (e.g. 'Auditor', 'Mumbai') recur across multiple table cells. Spatial coordinate keys prevent false cross-cell matching.

7. TP / FP / FN Definitions

- True Positive (TP): Pipeline prediction matching an accepted provisional annotation in type, coordinate key, and span.
- Known False Positive (FP): Pipeline prediction matching an explicitly rejected provisional candidate.
- Known False Negative (FN): Accepted provisional entity missed by the pipeline.
- Unassessed / Unknown: Predictions in unreviewed document regions (3,444 candidates, 5,712 predictions). These are tracked separately and NOT counted as false positives.

8. Provisional Annotation-Agreement Metrics

CRITICAL METRIC LABELING WARNING

These metrics measure agreement with the current automated provisional annotation subset (63 candidates). They are NOT independently human-validated model performance metrics. Zero candidates were reviewed by a human auditor.

Metric	Provisional Value	Evaluated Scope & Calculation
Precision	16.36%	$TP / (TP + FP) = 9 / (9 + 46) = 9 / 55$
Recall	60.00%	$TP / (TP + FN) = 9 / (9 + 6) = 9 / 15$
F1 Score	25.71%	$2TP / (2TP + FP + FN) = 18 / (18 + 46 + 6) = 18 / 70$
Entity Accuracy	N/A	Insufficient validated negative population

9. Accuracy

Entity-level accuracy is reported as N/A.

Justification: Entity-level accuracy requires a well-defined, independently validated true-negative population (a complete verified count of

non-PII spans across the entire document). Because 3,444 of 3,507 candidates remain unassessed, no true-negative denominator can be established without fabricating data.

10. Final / Human-Validated Metrics

Human-Validated Metric	Final Value	Reason Statement
Precision	N/A	No candidates were independently human-reviewed
Recall	N/A	No candidates were independently human-reviewed
F1 Score	N/A	No candidates were independently human-reviewed
Accuracy	N/A	No candidates were independently human-reviewed

11. Redaction Replacement Validation

Document-wide residual PII validation was conducted using scripts/residual_pii_checker.py across paragraphs, tables, headers, and footers:

- Provisional Annotation Slots Checked (source-chunk scope): 19
- Successfully Replaced (source-chunk scope): 19
- Residual Candidate Occurrences: 0
- Actual Residual PII: 0
- Key Entity Check: CareEdge Research (7 original occurrences -> 0 remaining occurrences -> 0 residual)
- Key Entity Check: Sarthak Malvadkar (7 original occurrences -> 0 residual)
- Key Entity Check: cs.connect@kshinternational.com (3 original occurrences -> 0 residual)

VALIDATION SCOPE DISCLAIMER

The '19 / 19 successfully replaced' figure measures redaction replacement efficacy on the currently checked candidate occurrences in their source annotated chunks. It must NOT be described as detector recall, model recall, PII detection accuracy, or 100% detection.

12. Regression Testing

The automated pytest suite verifies pipeline stability and bug fixes:

- Suite Result: 83 passed, 0 failed, 0 skipped
- Test Coverage: Includes 18 test modules covering detector unit tests, OpenXML table extractor logic, run-level redactor behavior, pseudonymizer seeding, evaluation metrics, and regression tests (test_careedge_leak_regression).

13. Limitations

1. Provisional Ground Truth: Ground truth was generated via policy simulation; 0 candidates were independently human-reviewed.
2. Partial Workload Assessment: Only 63 of 3,507 candidates (1.8%) have been assessed, leaving 3,444 unreviewed.
3. Statistical NER Model Constraints: en_core_web_sm is optimized for execution speed. Multi-word legal names in dense table cells may produce false positives or false negatives.
4. Single-Document Benchmark: Evaluation was performed on target prospectus documents; results should not be generalized as universal performance.
5. Public Repository Exclusion: The original prospectus (input/Red Herring Prospectus.docx) and raw annotation datasets (evaluation/ground_truth.json, ground_truth_verified.json) are intentionally excluded from the public GitHub repository to protect source data confidentiality.

14. Conclusion

The PII Redaction Tool implementation has been fully constructed, tested, and validated:

- The 7-stage OpenXML redaction engine detects PII across 9 supported categories and performs run-level text substitution.
- All 83 automated unit, integration, and regression tests pass cleanly (83 passed).
- Document-wide residual PII validation confirmed 0 actual residual PII among checked candidate occurrences, resolving the CareEdge Research propagation issue.
- Provisional annotation-agreement metrics against the 63-candidate reviewed subset are documented at 16.36% Precision, 60.00% Recall, and 25.71% F1.
- Final independently validated benchmark metrics remain N/A because zero candidates were independently human-reviewed.